

REDES NEURAIS COM TENSORFLOW



# PERCEPTRON DE ROSENBLATT

DIEGO RODRIGUES DSC

INFNET

# CRONOGRAMA

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# PERCEPTRON DE ROSENBLATT

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- BIBLIOTECAS
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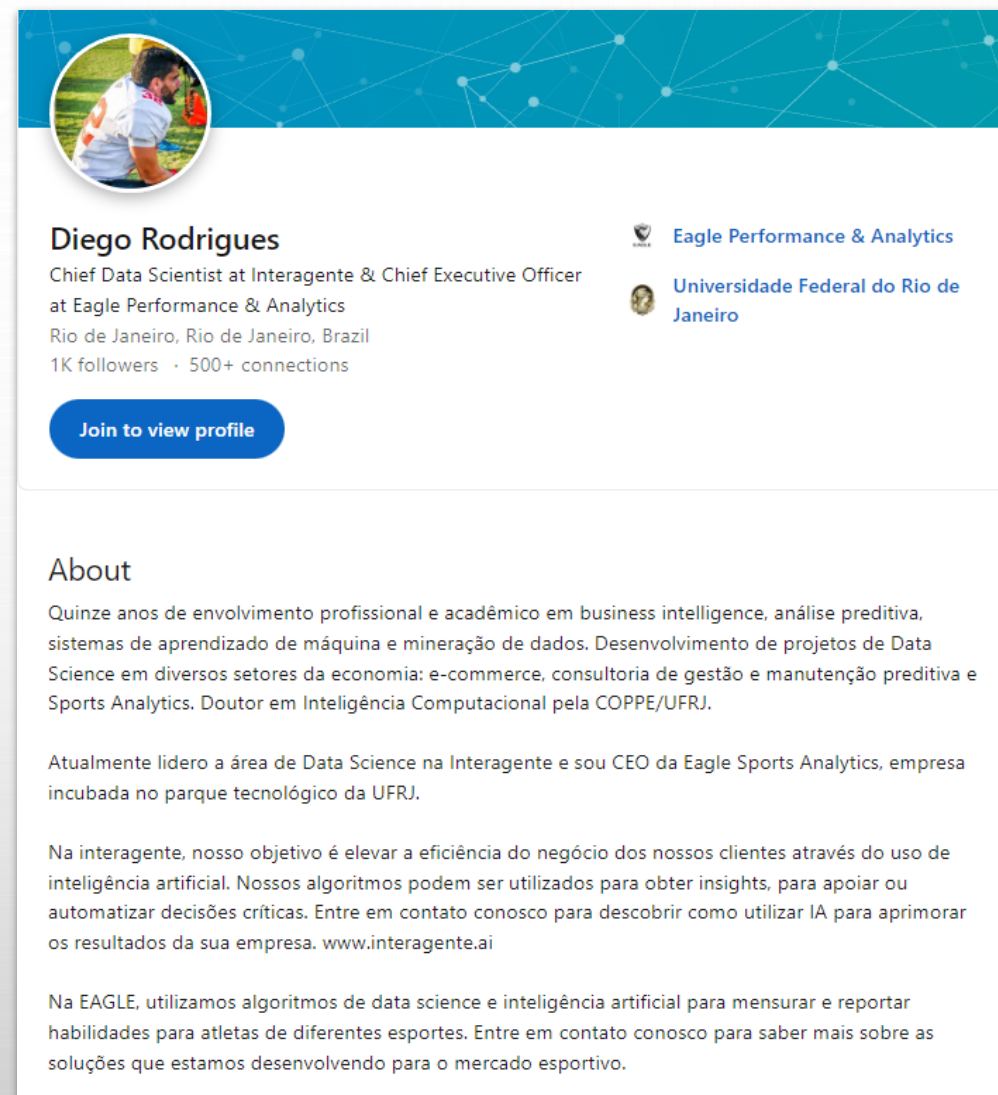
- PARTE 3 : TRABALHOS

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# PARTE 1 : TEORIA

# PROFESSOR DIEGO

- Doutor em Inteligência Computacional COPPE / UFRJ
- 16 anos de experiência com ciência de dados aplicada à engenharia.
- Sócio da Interagente e Eagle Sports Analytics.
- Professor da INFNET na Graduação, Pós-Graduação, Trilhas e Bootcamp.



The image shows a LinkedIn profile card for Diego Rodrigues. At the top is a circular profile picture of a man in a white shirt. Below the picture, the name 'Diego Rodrigues' is displayed in bold. Underneath the name, his current roles are listed: 'Chief Data Scientist at Interagente & Chief Executive Officer at Eagle Performance & Analytics', followed by his location 'Rio de Janeiro, Rio de Janeiro, Brazil' and his network statistics '1K followers · 500+ connections'. To the right of the name, there are two logos with company names: 'Eagle Performance & Analytics' and 'Universidade Federal do Rio de Janeiro'. A blue button with the text 'Join to view profile' is located below the main text. The 'About' section follows, starting with the heading 'About' and a paragraph describing his 15 years of experience in business intelligence, predictive analysis, machine learning, and data mining across various sectors like e-commerce and consulting. It also mentions his PhD from COPPE/UFRJ. Two more paragraphs describe his current roles: as a leader in Data Science at Interagente and CEO of Eagle Sports Analytics, and as someone using AI algorithms to enhance athletic performance at EAGLE.

**Diego Rodrigues**  
Chief Data Scientist at Interagente & Chief Executive Officer  
at Eagle Performance & Analytics  
Rio de Janeiro, Rio de Janeiro, Brazil  
1K followers · 500+ connections

[Join to view profile](#)

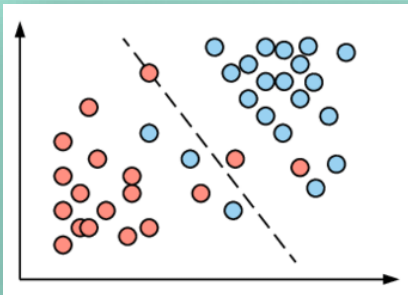
**About**  
Quinze anos de envolvimento profissional e acadêmico em business intelligence, análise preditiva, sistemas de aprendizado de máquina e mineração de dados. Desenvolvimento de projetos de Data Science em diversos setores da economia: e-commerce, consultoria de gestão e manutenção preditiva e Sports Analytics. Doutor em Inteligência Computacional pela COPPE/UFRJ.

Atualmente lidero a área de Data Science na Interagente e sou CEO da Eagle Sports Analytics, empresa incubada no parque tecnológico da UFRJ.

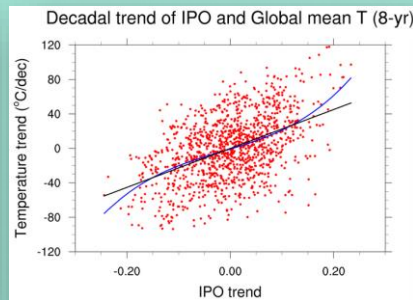
Na interagente, nosso objetivo é elevar a eficiência do negócio dos nossos clientes através do uso de inteligência artificial. Nossos algoritmos podem ser utilizados para obter insights, para apoiar ou automatizar decisões críticas. Entre em contato conosco para descobrir como utilizar IA para aprimorar os resultados da sua empresa. [www.interagente.ai](http://www.interagente.ai)

Na EAGLE, utilizamos algoritmos de data science e inteligência artificial para mensurar e reportar habilidades para atletas de diferentes esportes. Entre em contato conosco para saber mais sobre as soluções que estamos desenvolvendo para o mercado esportivo.

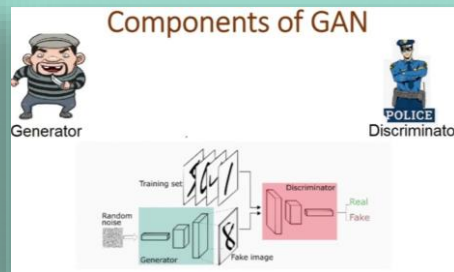
## APRENDIZADO SUPERVISIONADO



CLASSIFICAÇÃO

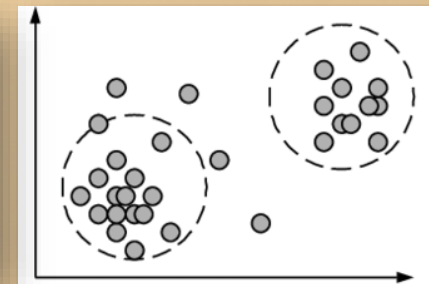


REGRESSÃO



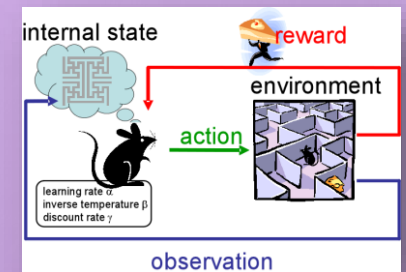
GENERATIVO

## APRENDIZADO NÃO- SUPERVISIONADO



AGRUPAMENTO

## APRENDIZADO POR REFORÇO



REFORÇO



# HISTÓRIA DAS REDES NEURAIS

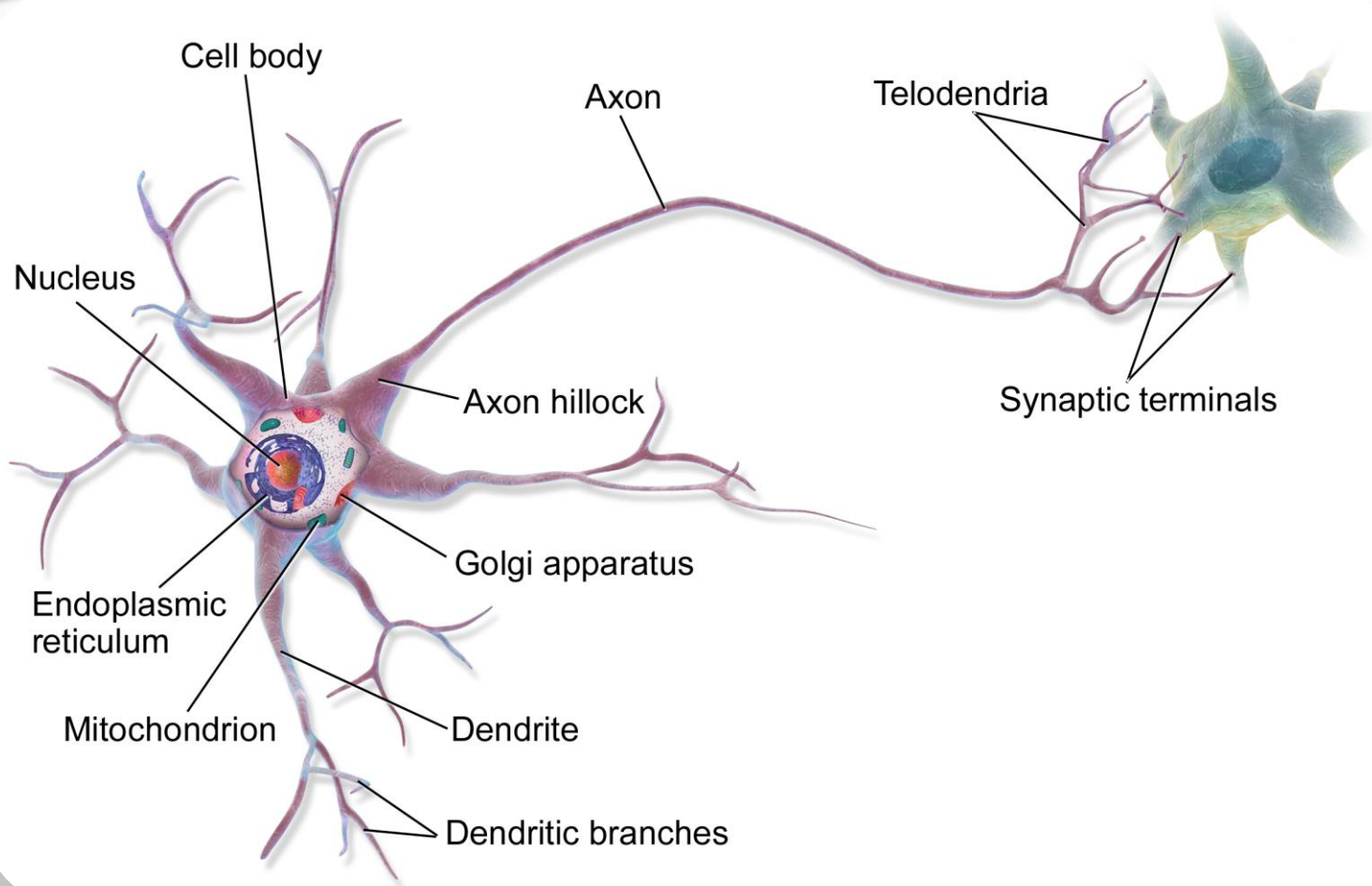
Perceptron, Rosenblatt 1954

Neocognitron, Fukushima 1979

Backpropagation Rumelhart, Hinton & Williams  
1986

LSTM, Hochreiter e Schmidhuber 1995

Keras & Tensorflow, Google 2015



# INSPIRAÇÃO BIOLÓGICA

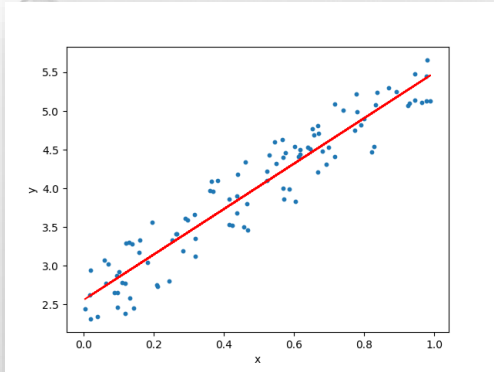


# O APROXIMADOR UNIVERSAL

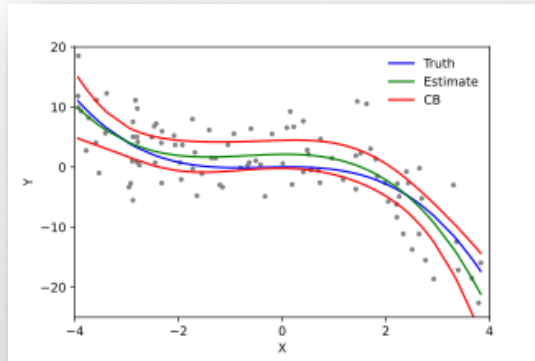
$$Y = \underbrace{F(X)}_{\text{Parte Determinística}} + \underbrace{\varepsilon}_{\text{Parte Estocástica}}$$

Parte Determinística

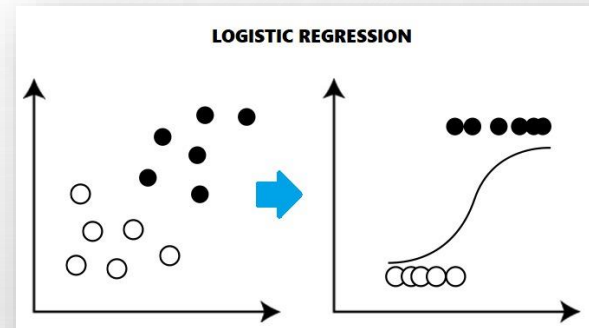
Parte Estocástica



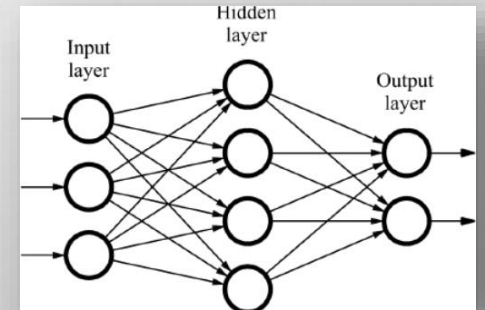
$$Y = \alpha^T x + \varepsilon$$



$$Y = X\alpha + \varepsilon$$

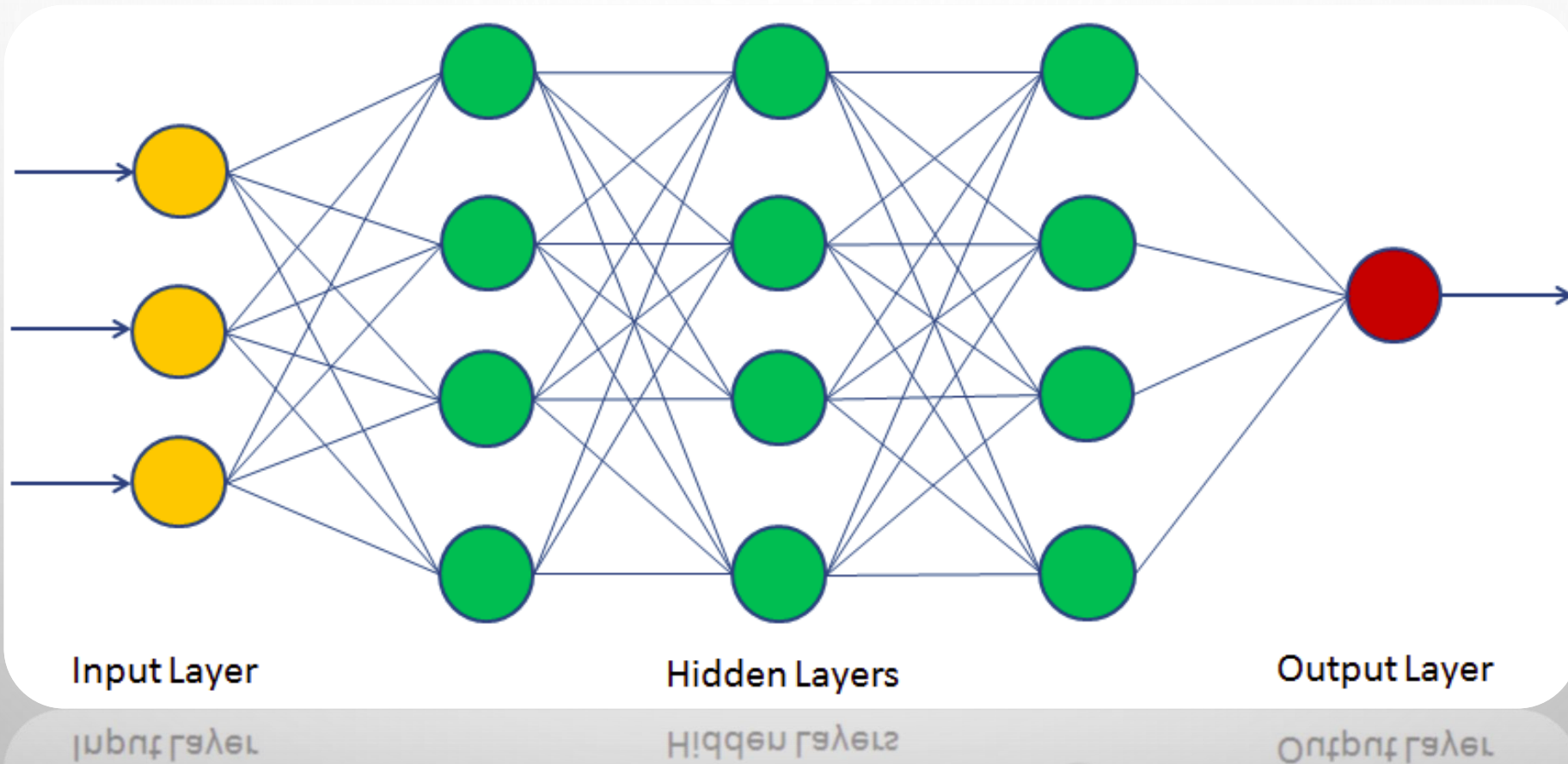


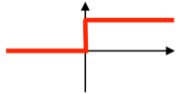
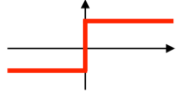
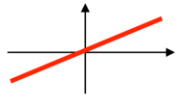
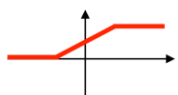
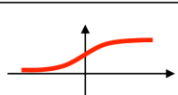
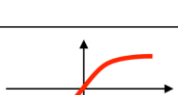
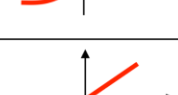
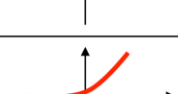
$$Y = \frac{1}{1 + e^{\alpha^T x + \varepsilon}}$$



$$Y = \varphi(x) + \varepsilon$$

# ○ APROXIMADOR UNIVERSAL



| Activation function                           | Equation                                                                                                                                          | Example                                   | 1D Graph                                                                             |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------|
| Unit step<br>(Heaviside)                      | $\phi(z) = \begin{cases} 0, & z < 0, \\ 0.5, & z = 0, \\ 1, & z > 0, \end{cases}$                                                                 | Perceptron<br>variant                     |    |
| Sign (Signum)                                 | $\phi(z) = \begin{cases} -1, & z < 0, \\ 0, & z = 0, \\ 1, & z > 0, \end{cases}$                                                                  | Perceptron<br>variant                     |    |
| Linear                                        | $\phi(z) = z$                                                                                                                                     | Adaline, linear<br>regression             |    |
| Piece-wise linear                             | $\phi(z) = \begin{cases} 1, & z \geq \frac{1}{2}, \\ z + \frac{1}{2}, & -\frac{1}{2} < z < \frac{1}{2}, \\ 0, & z \leq -\frac{1}{2}, \end{cases}$ | Support vector<br>machine                 |    |
| Logistic (sigmoid)                            | $\phi(z) = \frac{1}{1 + e^{-z}}$                                                                                                                  | Logistic<br>regression,<br>Multi-layer NN |    |
| Hyperbolic tangent                            | $\phi(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$                                                                                                     | Multi-layer<br>Neural<br>Networks         |    |
| Rectifier, ReLU<br>(Rectified Linear<br>Unit) | $\phi(z) = \max(0, z)$                                                                                                                            | Multi-layer<br>Neural<br>Networks         |   |
| Rectifier, softplus                           | $\phi(z) = \ln(1 + e^z)$                                                                                                                          | Multi-layer<br>Neural<br>Networks         |  |

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(<http://sebastianraschka.com>)

# FUNÇÕES DE ATIVAÇÃO

The image features a light gray background with a subtle radial gradient. In the corners, there are several realistic water droplets of varying sizes, some with highlights and shadows, giving them a three-dimensional appearance. The text is centered in a bold, black, sans-serif font.

# GOOGLE TENSORFLOW PLAYGROUND

The background is a light gray gradient. In the top-left and bottom-right corners, there are several realistic water droplets of various sizes, some overlapping. A faint, circular watermark is visible in the upper center of the page.

## PARTE 2 : PRÁTICA

# AMBIENTE DE DESENVOLVIMENTO

## Miniconda

Miniconda is a free minimal installer for conda. It is a small bootstrap version of Anaconda that includes only conda, Python, the packages they both depend on, and a small number of other useful packages (like pip, zlib, and a few others). If you need more packages, use the `conda install` command to install from thousands of packages available by default in Anaconda's public repo, or from other channels, like conda-forge or bioconda.

**Is Miniconda the right conda install for you?** The [Anaconda or Miniconda](#) page lists some reasons why you might want one installation over the other.

[System requirements](#)

[Latest Miniconda installer links by Python version](#)

[Installing Miniconda](#)

[Miniconda release notes](#)

[Other resources](#)

[Miniconda hash information](#)

## Simple Python Version Management: pyenv

[gitter](#) [join chat](#)

pyenv lets you easily switch between multiple versions of Python. It's simple, unobtrusive, and follows the UNIX tradition of single-purpose tools that do one thing well.

This project was forked from [rbenv](#) and [ruby-build](#), and modified for Python.

```
$ pyenv versions
2.7.10
* 3.5.0 (set by /Users/uyu/.pyenv/version)
miniconda3-3.16.0
pypy-2.6.0

$ python --version
Python 3.5.0

$ pyenv global pypy-2.6.0

$ python --version
Python 2.7.9 (295ee98b69288471b0fcf2e0ede82ce5209eb90b, Jun 01 2015, 17:30:13)
[PyPy 2.6.0 with GCC 4.9.2]

$ cd /Volumes/treasuredata/jupyter

$ pyenv version
miniconda3-3.16.0 (set by /Volumes/treasuredata/.python-version)

$ python --version
Python 3.4.3 :: Continuum Analytics, Inc.
```

Boa Prática: 1 Ambiente por projeto!



# CRIANDO AMBIENTE COM CONDA

## Managing environments

With conda, you can create, export, list, remove, and update environments that have different versions of Python and/or packages installed in them. Switching or moving between environments is called activating the environment. You can also share an environment file.

There are many options available for the commands described on this page. For a detailed reference on all available commands, see [commands](#).

## Creating an environment with commands

Use the terminal for the following steps:

1. To create an environment:

```
conda create --name <my-env>
```

Replace `<my-env>` with the name of your environment.

2. When conda asks you to proceed, type `y`:

```
proceed ([y]/n)?
```

This creates the myenv environment in `/envs/`. No packages will be installed in this environment.

3. To create an environment with a specific version of Python:

```
conda create -n myenv python=3.9
```

## Activating an environment

Activating environments is essential to making the software in the environments work well. Activation entails two primary functions: adding entries to PATH for the environment and running any activation scripts that the environment may contain. These activation scripts are how packages can set arbitrary environment variables that may be necessary for their operation. You can also [use the config API to set environment variables](#).

Activation prepends to PATH. This only takes effect when you have the environment active so it is local to a terminal session, not global.

### Note

When [installing Anaconda](#), you have the option to "Add Anaconda to my PATH environment variable." *This is not recommended* because it *appends* Anaconda to PATH. When the installer appends to PATH, it does not call the activation scripts.

### Note

On Windows, PATH is composed of two parts, the *system* PATH and the *user* PATH. The system PATH always comes first. When you install Anaconda for "Just Me", we add it to the *user* PATH. When you install for "All Users", we add it to the *system* PATH. In the former case, you can end up with system PATH values taking precedence over your entries. In the latter case, you do not. *We do not recommend* [multi-user installs](#).

To activate an environment: `conda activate myenv`

# AMBIENTE PYTHON



4. Variáveis  
Aleatórias



1. Editor de Código

5. Visualização



2. Gestor de Ambiente



6. Machine  
Learning

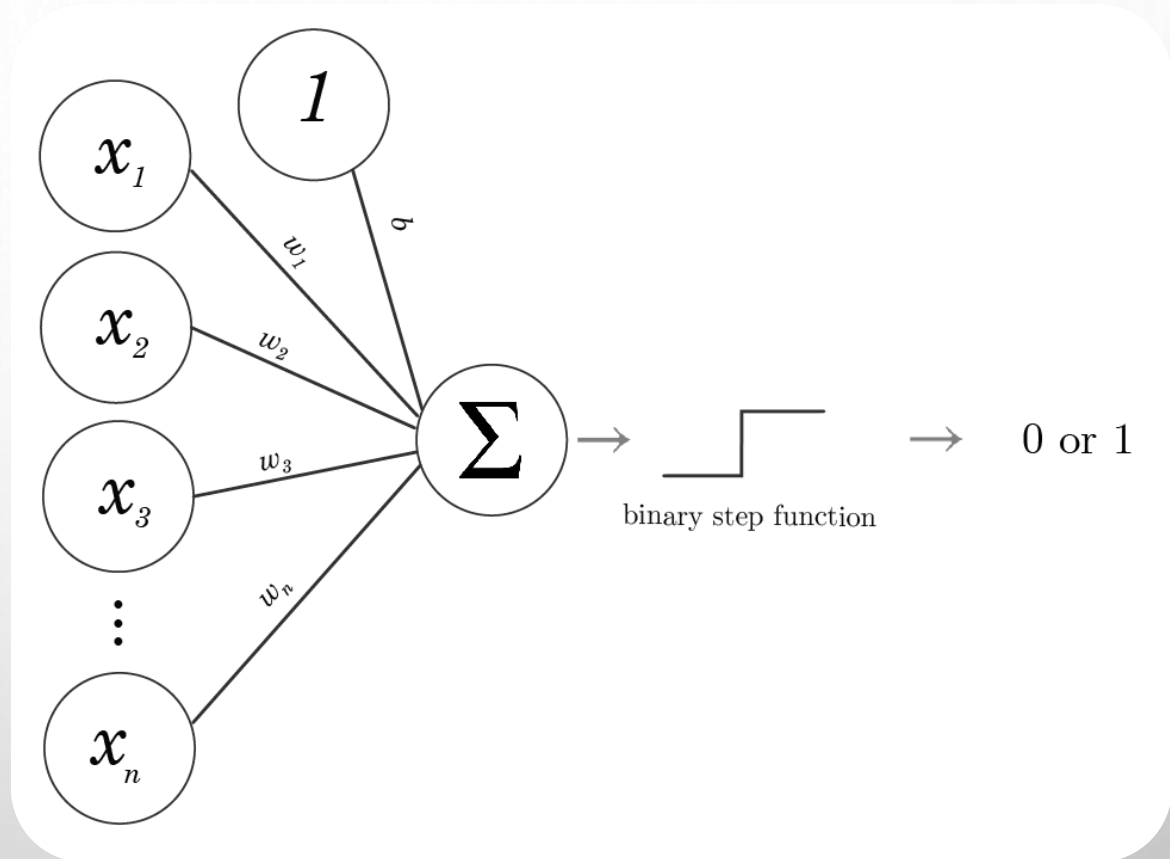


3. Ambiente  
Python do Projeto



3. Notebook  
Dinâmico

# PERCEPTRON DE ROSENBLATT





# EXERCÍCIO: PERCEPTRON DE ROSENBLATT

## PRÓXIMA AULA: CLASSIFICAÇÃO COM NEURÔNIO SIGMÓIDE

